

Where Our Beds Are Under Pressure

What two years of operational data reveals, and what to do about it

Albion Care Network | 5 hospitals | 8 wards | 760 beds | 131,000 admissions

The business objective

Replace fragmented spreadsheets and monthly averages with a single operational view of bed occupancy, so managers can see where pressure sits today and act on it.

What we set out to deliver

- One trusted source across five hospitals
-
- Ward-level pressure visible at a glance each morning
-
- Evidence of what actually drives occupancy
-
- A clear view of where staffing and demand diverge

5

Source systems unified

760

Beds monitored

2

Dashboards delivered

731

Days of history

Before this, no single view existed. Occupancy, admissions and staffing lived in separate systems.

We manage beds on averages. Pressure does not average out.

71.9%

Average network occupancy

100%

Peak hit on every inpatient ward

48%

Of the year Orthopaedics B is effectively full

What this costs us today

- Capacity problems are discovered after they bite, never before
-
- Safe staffing ratios are breached on 4.3% of ward-shifts, rising to 6.4% in December
-
- Elective surgery absorbs the pressure through late cancellation, reaching 11.9% in winter
-
- A small group of long-stay patients quietly consumes a quarter of all bed capacity

Four findings from two years of data

1

Emergency admissions drive occupancy

Correlation of 0.62 with daily occupancy, ahead of every other factor. Emergencies are 79.9% of all admissions.

2

Elective surgery does not drive it

Correlation of only 0.13. Schedules are fixed months ahead, so bed pressure is absorbed through cancellation instead.

3

Staffing does not follow demand

Correlation of -0.20. February has the highest occupancy of the year and the lowest staffing.

4

A few long stays consume the capacity

Stays beyond seven days are 8.7% of admissions but occupy 26.4% of all bed-hours.

The pattern is consistent and repeats across both years. That means these are structural problems, not bad luck.

Operations: where to act today



Available Beds

165

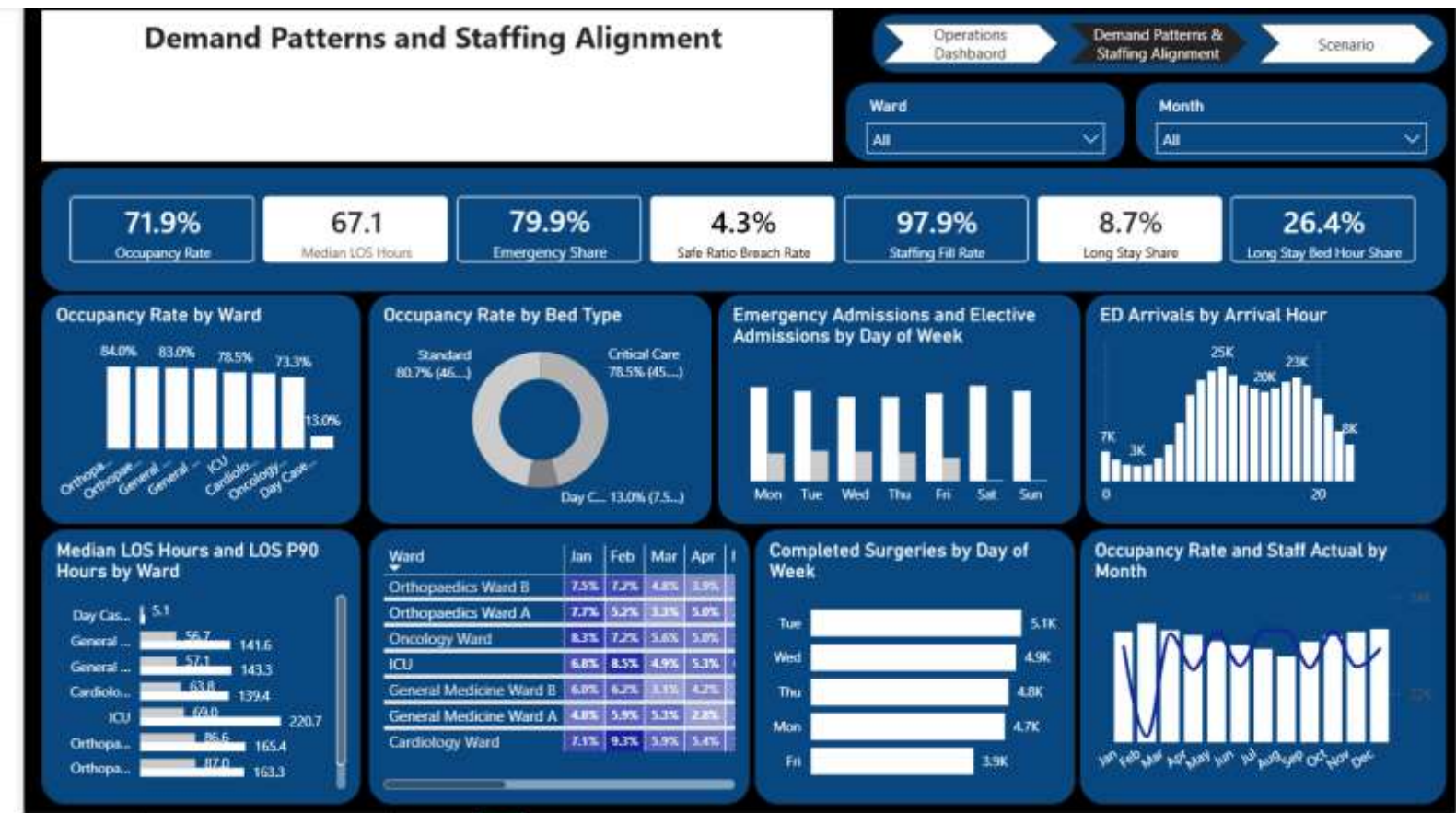
of 760 across the network

What it gives a bed manager

Wards ranked by pressure, not alphabetically, so the top row is where to go first.

Four wards sit under severe pressure. Patient flow shows whether beds are filling or emptying.

Demand and staffing: why pressure happens



February
Busiest month at 81.6% occupancy

Lowest staffing of the year

Breach rate 6.2%, against 3.5% in summer

Three repeating cycles
Yearly: February peak, August trough
Weekly: electives weekdays only
Daily: ED peaks late morning

The heatmap shows breaches clustering in exactly the months wards are fullest.

Our busiest months are our most thinly staffed

Month	Occupancy	Staffing level	Safe ratio breaches
February	81.6%	Lowest	6.2%
December	78.0%	Low	6.4%
June	66.8%	Average	3.5%
August	59.0%	Highest	3.5%

Rotas run on fixed patterns. Demand swings 23 percentage points across the year. Closing that gap is the single clearest opportunity.

Four actions, ranked by impact

1

Review long stays daily

Target patients approaching day seven. They are 8.7% of admissions but hold 26.4% of bed-hours, so this releases more capacity than anything else available.

Low effort · High impact

2

Rebalance winter staffing

Shift establishment toward December to February, when occupancy peaks and safe-ratio breaches nearly double.

Medium effort · High impact

3

Move electives into August

Occupancy falls to 59% in August while cancellations peak at 11.9% in December. Rebalancing uses capacity that currently sits idle.

Medium effort · High impact

4

Set a 95% escalation trigger

Wards spend 26% to 48% of the year at or above this level. The trigger should drive a defined staffing response, not just an alert.

Low effort · Medium impact

From dashboard to daily practice

Weeks 1 to 2	Put the dashboard in the morning huddle Bed managers open the operations page at 9am. The ward pressure ranking becomes the agenda.	Bed managers
Weeks 3 to 4	Start the daily long-stay review A list of patients approaching day seven, with a named owner for each discharge barrier.	Ward leads, discharge team
Weeks 5 to 8	Build the next rota from the data Use monthly occupancy patterns to set winter establishment. Track breaches weekly to prove the effect.	Workforce planning
Weeks 9 to 12	Pilot the elective shift Move one specialty's activity from the December peak into August. Measure the cancellation rate.	Scheduling, theatres
Ongoing	Review pressure monthly Track the four severe-pressure wards and the safe-ratio breach rate as standing board metrics.	Executive, operations

Three decisions today

1

Approve the 95% escalation threshold

Confirm with clinical leadership that 95% occupancy is the right trigger for a defined staffing response.

2

Name an owner for the long-stay review

The highest-impact recommendation needs a named person and a daily slot to become real.

3

Approve the elective pilot

One specialty moving a tranche of activity into August would prove the case at low risk.

Limitations we are open about

Correlation does not prove causation; the staffing finding is consistent with fixed rostering but needs rota policy to confirm. ICU Step-down bed capacity is missing from the source data, so pressure is slightly understated. Data runs to 31 December 2025.